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When Color Contrast Is Not Enough: An Empirical Study of Dynamic Color Adaptation for In-game Health Visualizations

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Abstract

We present a dynamic color assignment approach and an empirical study of context-adaptive color for fast-moving game visualizations. In first-person shooter (FPS) games, players often need to read visualizations that are attached to moving characters while making quick decisions. Health visualizations are particularly important because they help players identify targets and decide when to act. As characters move across changing backgrounds, maintaining sufficient contrast becomes a natural design problem. We implemented a real-time, background-adaptive color assignment approach in *RobotLife*, an open-source FPS game. Our approach samples the local background around a health visualization and dynamically adjusts its color to meet a WCAG-aligned contrast ratio. We evaluated our approach in a within-subjects study with 24 participants, comparing dynamic color assignment conditions and fixed color baselines across three health visualization designs. Our results show that, in *RobotLife*, dynamic-color assignment did not significantly improve readability or aesthetics for in-game visualizations under motion. Instead, participants prioritized glanceable readability and semantic consistency, revealing trade-offs among readability, aesthetics, and semantic stability between dynamic color assignment and the fixed-color baseline. All supplemental materials of our work are available at `osf.io/csjrb/`.

Index Terms: Situated visualization, Video games

1 Introduction

Data visualization helps players understand game states and make quick decisions, especially in fast-paced games such as first-person shooters (FPS). Prior work shows that in-game visualization design is highly context-dependent, with different genres and mechanics requiring different visual forms, layouts, and interaction strategies [4, 33]. HUD studies further suggest that players often prefer familiar interfaces over novel alternatives, even when alternatives may offer design advantages [25, 9]. Thus, effective in-game visualizations depend not only on visual clarity, but also on gameplay context, player habits, and established conventions.

Yao *et al.*'s work on in-game visualizations [32] is closest to ours. They examined how different ways of embedding health visualizations into moving game characters affect user experience, comparing three designs with different levels of visual integration: Non-Integrated, Fully-Integrated, and Partially-Matched. Their findings revealed trade-offs between readability, aesthetics, and integration with the game scene: a visualization that blends well with a character may improve visual coherence but become harder to read in motion. Building on their findings, our study examines whether dynamic color adaptation can improve the readability of in-game visualizations as the surrounding game environment changes.

Color contrast is a natural concern in FPS games, where players must read health visualizations on moving characters while tracking targets, navigating space, and making rapid decisions. As characters move through complex environments, the background behind a health visualization continuously changes. A fixed color that is readable against one background may become less visible against another, raising a practical design question: can dynamic color adaptation improve the readability and perceived quality of in-game health visualizations under motion?

To answer this question, we implemented a real-time, background-adaptive color assignment approach in *RobotLife* [3], an open-source FPS game. Our approach samples the local background around a health visualization and dynamically adjusts its color to meet a WCAG-aligned contrast ratio [26]. We then conducted a within-subjects study with 24 participants, comparing dynamic color assignment and fixed-color baselines across the three health visualization designs from Yao *et al.* [32]. Our work empirically examines how dynamic color assignment affects the readability, aesthetics, and semantic consistency of in-game health visualizations under motion in a tested FPS game.

2 Related Work

Our work examines dynamic color changes for in-game visualizations attached to and moving with data referents (*i.e.*, game characters). We thus review situated and embedded visualization, color assignment, and in-game visualizations.

2.1 Situated and Embedded Visualization

Situated visualization integrates visual representations with the spaces, objects, and activities of a physical environment. Bressa *et al.* [5] surveyed its definitions and related work, with examples spanning mobile devices [29, 22, 14, 16] and Augmented Reality (AR) [35, 1, 2, 6, 8]. Building on this area, Willett *et al.* [30] defined embedded visualization as visualizations placed on or near their data referents, reducing cognitive distance caused by spatial or temporal separation. Yao *et al.* [31] further proposed *visualization in motion*, referring to relative movement between the viewer and the visualization. Their studies showed that people can still read information from visualizations moving at high speeds and along irregular trajectories. Our work aligns with this definition, as FPS games involve continuous relative movement between the seated player and moving game characters with embedded visualizations.

2.2 Color Assignments in Visualization

Visualization researchers have developed many color-assignment strategies to support category differentiation. Wang *et al.* [28] selected visible color assignments from a predefined library by considering the spatial distribution of data points, while Lu *et al.*'s *Palettailor* [18] optimized palettes for multi-class charts based on category overlap and density. Salvi *et al.*'s *ColorMaker* [21] further used simulated annealing to optimize mappings based on user-

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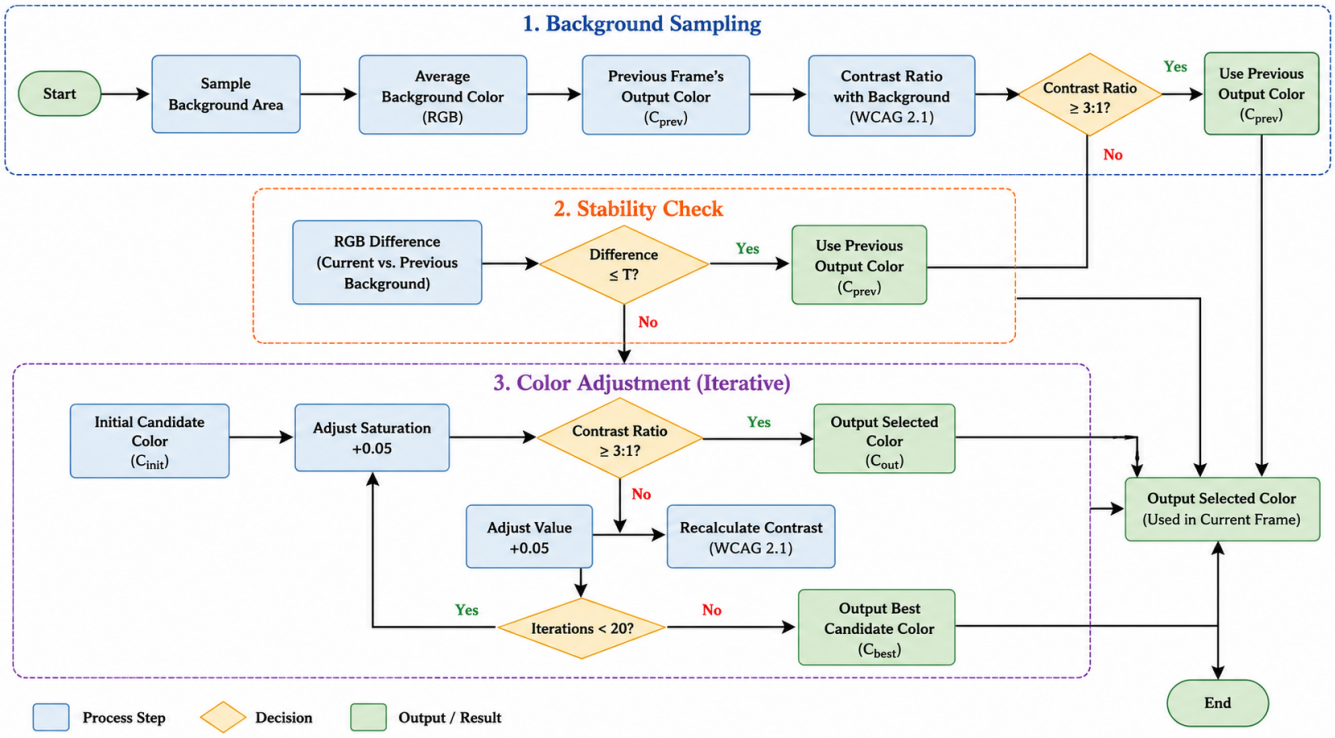


Figure 1: An overview of our dynamic color assignment approach. The entire pipeline includes spatial sampling, frame comparison with a stability check, and an iterative color adjustment loop.

specified base colors and perceptual principles. Beyond discriminability, color can also preserve semantic meaning and cross-view consistency, as in *TIARA* [17], which maintains topic colors across multiple views, and *EvoRiver* [24], which uses meaningful hues to distinguish topic relations and leader categories. Color assignment has also been studied for dynamic and temporal data. *Cuttlefish* [27] dynamically allocates, compresses, and recycles colors for hierarchical data during pan and zoom, while Chen *et al*. [10] adapted sub-category colors based on parent hues. In temporal visualizations, stable colors help users track changes over time, as shown in dynamic word clouds and coordinated trend views [11] and *RankExplorer* [23]. Other work has optimized or locally remapped colormaps to emphasize spatial patterns or support visual search in dense data [34, 13]. Together, prior work shows that color assignment can support discrimination, semantic interpretation, cross-view correspondence, temporal continuity, and spatial pattern enhancement. However, these techniques are typically designed for static visualizations, interactive tools, or dynamic data under relatively controlled visual contexts. Our work extends this direction to FPS game environments, where both the visualization and its background continuously change.

2.3 Visualizations in Video Games

Visualizations in video games support task completion, decision-making, and immersion, but their design is highly context-dependent. Bowman *et al*. [4] proposed a five-dimensional framework for analyzing game visualizations across genres, covering purpose, audience, temporal use, visual complexity, and immersion. Zammitto *et al*. [33] similarly showed that effective visualization tactics depend on game genre and player tasks. Beyond player-facing interfaces, visualizations have also been used to analyze gameplay data, such as overlaying player behavior on game maps to reveal spatial patterns [19]. Prior studies have examined how players perceive and use in-game visualizations. El-Nasr *et al*. [12] found that task goals guide visual attention, with color and motion playing secondary roles. In FPS games, players' gaze often centers on the screen, making placement and readability especially important. HUD studies further show that effectiveness depends on task and familiarity: Caroux *et al*. [9] found that experienced players are more sensitive to HUD changes and often prefer familiar layouts, while standard game HUDs can match or outperform professional designs for target identification and subjective preference [25]. Peacocke *et al*. [20] further showed that no display type is universally superior across HUDs, diegetic displays, and spatial representations. Closest to our work, Yao *et al*. [32] compared three in-game visualization designs with different levels of integration. Their results showed design-specific trade-offs across motion contexts, highlighting the need to balance readability, aesthetics, and integration with the game scene. In contrast, our work focuses on dynamic color assignment, using video games as a motion-based testbed to examine whether real-time, background-adaptive color can improve readability and user preference as both the visualization and background continuously change.

3 Dynamic Color Assignment Approach

We present our dynamic color assignment approach first, followed by its implementation. Because our study focuses on color adaptation between a visualization and its changing visual background, we describe the approach in the context of *RobotLife* [3], an FPS game with busy and dynamic scenes.

3.1 Approach Design

We designed our dynamic color assignment approach to evaluate whether real-time, background-adaptive color can improve the readability of health visualizations in motion. In FPS games, both

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the visualization and its surrounding background change continuously as characters move through the scene. Our approach therefore adapts the health visualization color to its immediate visual context, aiming to maintain sufficient contrast while avoiding unnecessary changes that could reduce recognizability or cause visual distraction. As illustrated in Figure 1, our approach consists of three steps: **background sampling**, **contrast and stability checking**, and **iterative color adjustment**, together determining when a color update is needed and how the output color should be adjusted.

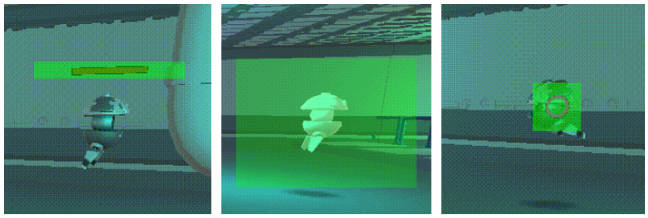


Figure 2: Sampling regions for the three visualization designs. Green rectangles mark the local background areas sampled by our dynamic color assignment approach for the Non-Integrated (left), Fully-Integrated (middle), and Partially-Matched (right) designs.

Step 1: Background Sampling. We first sample the local background around the health visualization, since readability depends on the immediate visual context rather than the global scene color. For the Non-Integrated and Partially-Matched designs, which are screen-space UI elements, we define the sampling region S as a rectangle centered at the UI coordinates (x, y). For the Fully-Integrated design, which is attached to the robot model, we anchor S to the model's world-space pivot point and project it onto the screen. For screen-space visualizations, we set S to $2.0\times$ the visualization area; for the Fully-Integrated design, we use a $1.5\times$ scale around the projected screen-space extent of the rendered body region. These regions capture nearby contextual colors that may affect perception while avoiding unrelated scene areas. We then compute the average background color C_{bg} from all pixels in S:

$$C_{bg} = \frac{1}{n}\sum_{i=1}^{n} P_i \tag{1}$$

where P_i is the RGB vector of the i-th pixel in S, and n is the number of sampled pixels. The mean color provides a compact representation of the local background for efficient real-time adaptation.

Step 2: Contrast and Stability Checking. After obtaining the current background color, we check whether the previous output color C_{t-1} still provides sufficient contrast. This prevents frame-by-frame color changes that could cause flickering or visual distraction. We compute the contrast ratio (CR) between C_{t-1} and C_{bg} using the WCAG 2.1 [26] definition:

$$CR = \frac{L_1 + 0.05}{L_2 + 0.05} \tag{2}$$

where L_1 and L_2 are the relative luminance values of the lighter and darker colors, respectively. Following the WCAG 2.1 Level AA non-text criterion [26], we use $3:1$ as the target contrast ratio for graphical objects and UI components. Since health bars are graphical UI elements, this threshold provides a suitable baseline for assessing their distinguishability from the background.

If $CR \geq 3:1$, we preserve C_{t-1} as the current output colorto maintain temporal stability and avoid unnecessary updates while the color remains readable. If $CR < 3:1$, we compare the current and previous input colors and trigger adjustment only when at least one RGB channel differs by more than 0.2. We used this RGB difference threshold as an empirical compromise between responsiveness and temporal stability: smaller values overreacted to texture noise, lighting changes, or camera movement, while larger values delayed meaningful adaptation.

Step 3: Iterative Color Adjustment. When a color update is required, we adjust the visualization color in HSV space, which separates hue, saturation, and value for more direct perceptual control than RGB. We keep hue fixed to preserve the semantic identity of the health visualization, as players often associate colors with game meanings, such as red with health or danger. We then adjust saturation and value to improve contrast while maintaining semantic continuity. We prioritize saturation adjustment because it can improve contrast while preserving hue and semantic identity. Saturation is increased in steps of 0.05 to balance gradual change and convergence speed, with a cap of 20 iterations to cover the full normalized range in the worst case ($20 \times 0.05 = 1.0$):

$$S_{n+1} = \min(S_n + 0.05, 1.0) \tag{3}$$

After each update, we convert the adjusted HSV color back to RGB and recompute its contrast ratio with C_{bg}. If the contrast ratio reaches $3:1$, the loop terminates and the adjusted color is used for the current frame. If saturation alone is insufficient, we then adjust value, increasing brightness to improve contrast against darker or complex backgrounds. As with saturation, value is updated for up to 20 iterations in steps of 0.05:

$$V_{n+1} = \min(V_n + 0.05, 1.0) \tag{4}$$

The algorithm recomputes contrast after each iteration and stops once the threshold is reached. The final adjusted color is used as C_t and stored for comparison in the next frame. As shown in Figure 3, this procedure lets health visualizations respond to changing backgrounds while limiting unnecessary color changes, balancing contrast, temporal stability, and semantic recognizability in real time.

3.2 Approach Implementation

We implemented dynamic color adjustment for both screen-space 2D health bars and world-space 3D shader-rendered models. For screen-space health bars, we sampled a predefined screen region using `Texture2D.ReadPixels`, computed the average background color, generated a contrast color, and refined it with our custom `ColorContrastUtility` to meet the target contrast ratio. We use `Color.Lerp` for smoother transitions for the real-time health bar color updating, with sampling triggered periodically through Unity coroutines. For shader-rendered models, we used a secondary camera, `EnemyBackgroundSampler`, to render the environment behind the character into a `RenderTexture`. We projected the character's screen-space bounding box to define the sampling region, processed the sampled background color through the same contrast adjustment pipeline, and passed the final color to the shader via a `MaterialPropertyBlock`, enabling efficient updates without modifying shared materials. Both implementations used Unity's core libraries, including `UnityEngine` and `UnityEngine.UI`, together with our dynamic color assignment approach.

4 Empirical Study

To evaluate how dynamic color assignment affects the readability and perceived quality of in-game health visualizations, we conducted an empirical study in *RobotLife* [3]. Our study was IRB-approved (XJTLU, № ER-LRR-11000180720251112115655). Below, we describe the evaluated variables, apparatus, procedure, and data collection and analysis.

4.1 Evaluation Variable

Our measured variable is the color assignment condition of the health visualization: Vd dynamic color assignment, where the color adapts in real time to the local background, and Vf fixed color, where the visualization remains red throughout gameplay. In our

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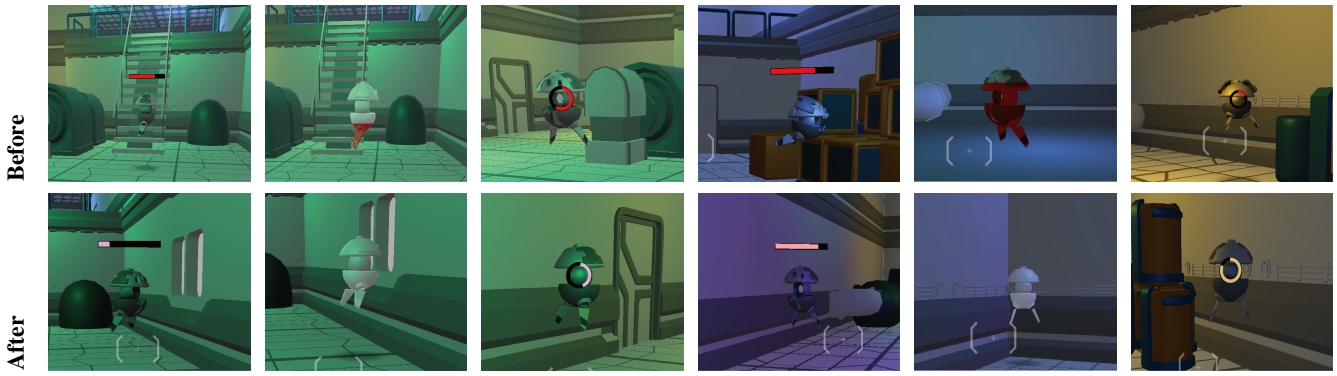


Figure 3: Examples of health visualization colors before and after dynamic color adjustment in *RobotLife*. The first row shows the original colors before algorithmic adjustment, while the second row shows the corresponding colors after adjustment under the same background conditions.

implementation, Vd may change hue in response to the sampled background, whereas Vf preserves a stable red-health mapping throughout gameplay. We evaluate both color conditions across three health visualization designs from Yao *et al*. [32]: A Non-Integrated, B Fully-Integrated, and C Partially-Matched, which represent different levels of integration with the game character. Thus, comparing Vd and Vf captures differences in adaptive versus fixed color assignment, temporal color behavior, and semantic color consistency. To reduce ordering effects, we counterbalance the three visualization designs using six Latin-square sequences: A B C, A C B, B A C, B C A, C A B, and C B A. Within each sequence, we counterbalance the two color orders: Vd Vf and Vf Vd, yielding $6 \times 2 = 12$ study orders. Each participant completed six conditions: three visualization designs crossed with two color assignment conditions.

4.2 Apparatus

We conducted our study in a controlled laboratory environment. All sessions were administered on a self-assembled desktop, equipped with the hardware specifications detailed in Table 1.

Component	Specification
Monitor	ASUS 24-inch (144Hz refresh rate)
System	Windows 11
CPU	Intel Core i5-12400KF
Graphics Card	AMD Radeon RX 6750 GRE (12GB VRAM)
Memory	32GB DDR4 (3200MHz)
Input	ASUS full-sized keyboard; Logitech G Pro Wireless mouse

Table 1: Hardware specifications used in our empirical study.

4.3 Procedure

Pre-questionnaire: Participants first signed an informed consent form. Those who agreed continued to a pre-questionnaire covering demographics (*i.e*., gender identification and age range) and gaming background. They self-rated their general gaming experience and PC keyboard-and-mouse proficiency on 7-point Likert scales, reported their general gaming, FPS gaming, and PC gaming frequency, and selected their preferred game genres (*e.g*., FPS, Platformers, Action/Adventure, RPG, TPS, Sports games, Puzzle games; multiple choices).

Tutorial: Participants then completed a guided tutorial introducing the health bar designs, game rules, and shooting actions.

Main Tasks: Each participant completed all six conditions. In each condition, they played for 5 min and were asked to locate and destroy only the eight hostile robots with health values $> 66\%$. The round restarted immediately if participants eliminated all eight targets or reached the failure threshold by destroying three friendly robots. Each condition still lasted 5 min, regardless of round completion. After each condition, participants rated readability and aesthetics on 7-point Likert scales (1 = Strongly disagree; 7 = Strongly agree). We used *PREVis* [7] (*Understand*, *Data Readability*, *Layout*, and *Data Features*) to assess perceived readability, and *BeauVis* [15] (*Enjoyable*, *Likable*, *Pleasing*, *Nice*, and *Appealing*) to measure visualization aesthetics. Both are validated scales in the visualization community.

Post-questionnaire: After main tasks, participants completed a post-questionnaire and a semi-structured interview. They ranked the visualization designs from 1st to 3rd based on *perceived readability*, *aesthetics in context*, *mission support*, and *overall gaming experience*. The interview collected open-ended feedback on the dynamic color assignment, visual fatigue or discomfort, and suggestions for improving the visualization design or color adaptation.

4.4 Data Collection and Analysis

We collected both quantitative and qualitative data. Quantitative data included participants' Likert-scale ratings, rankings of the three health visualization designs, and system-recorded gameplay logs. From the logs, we derived four combat-performance measures: *correct-hit efficiency*, the proportion of hits on enemy robots among all hit events; *correct-kill efficiency*, the proportion of kills on enemy robots among all kill events; *wrong-hit rate*, the proportion of hits on friendly robots among all hit events; and *wrong-kill rate*, the proportion of kills on friendly robots among all kill events. Because all participants completed all six conditions, we treated comparisons between Vf and Vd as paired observations. We summarized quantitative data using descriptive statistics, including means, medians, standard deviations, and rating distributions. For inferential analysis, we used paired-samples t-tests to compare Vf and Vd within each visualization design, and report Cohen's d as the paired effect size. We applied Bonferroni correction separately to the PREVis and BeauVis comparison families: across 12 PREVis comparisons (3 designs $\times$ 4 subscales) and 3 BeauVis comparisons, one per design. Qualitative data came from participants' open-ended interview responses. We audio-recorded and transcribed the interviews, then analyzed them using thematic analysis. One author generated initial codes, which were grouped into recurring themes and discussed among the authors to resolve ambiguities and refine interpretations.

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4.5 Contrast Audit

To quantify contrast behavior in our test environment, we conducted a route-based contrast audit for the three health visualization designs (A , B , C), under both Vf and Vd. We followed a fixed scripted route and recorded samples every 0.5 s, balancing temporal coverage with redundancy: denser sampling produced near-duplicate frame observations, while sparser sampling risked missing short-lived contrast failures. At each sampling moment, we captured the local background region for each visible enemy health visualization and computed the contrast ratio between the displayed health color and sampled background color. For Vf, the color remained fixed red; for Vd, it was assigned by our background-adaptive method. We also logged whether each sample met the WCAG-aligned 3:1 threshold. To exclude targets that were technically present but not meaningfully readable, we analyzed only visible samples, defined as health visualizations that were on screen and exceeded a minimum screen-space size threshold. For each design and color condition, we summarized the number of visible samples, failing samples, failure rate, mean contrast ratio, and mean contrast deficit relative to the 3:1 threshold.

5 Results

In this section, we present our participants' sample, then the quantitative results, and finally the qualitative results.

5.1 Participants

We recruited $N = 24$ participants through social media platforms, including $n = 21$ males and $n = 3$ females. Most participants ($N = 20$) were aged 25–30, while the remaining $N = 4$ were aged 18–24. Participants reported relatively high PC gaming experience. On 7-point Likert scales, they rated themselves as skilled gamers ($M = 4.71$, $SD = 1.57$) and confident with mouse-and-keyboard control ($M = 4.83$, $SD = 1.65$). Most participants played PC games daily (58.3%, $N = 14$) or weekly (33.3%, $N = 8$), and 75% ($N = 18$) reported a preference for FPS games. Overall, the sample consisted mainly of experienced PC gamers with relevant FPS familiarity.

5.2 Contrast Audit Results

We analyzed route-based contrast ratios for the three health visualization designs under both Vf and Vd. Table 2 reports visible samples, failing samples, failure rates, mean contrast ratios, and mean contrast deficits relative to the 3:1 threshold.

Table 2: Route-based contrast audit results for the fixed-red baseline (Vf) and dynamic-color condition (Vd) across the three health visualization designs. The table reports visible samples, failing samples, failure rate, mean dynamic contrast ratio, and mean dynamic contrast deficit using the 3:1 threshold.

Design	Color	Visible n	Fail n	Fail rate (%)	Mean contrast	Mean deficit
A Non-Integrated	Vf	899	882	98.11	1.64	1.36
B Fully-Integrated	Vf	653	587	89.89	2.09	0.91
C Partially-Matched	Vf	793	761	95.96	1.73	1.27
A Non-Integrated	Vd	602	140	23.26	3.83	0.17
B Fully-Integrated	Vd	675	185	27.41	4.19	0.29
C Partially-Matched	Vd	807	168	20.82	3.91	0.15

The fixed-red baseline frequently failed the 3:1 threshold, with failure rates of 98.11%, 89.89%, and 95.96% for A , B , and C , respectively. Its mean contrast ratios were also below threshold across all designs (1.64, 2.09, and 1.73). In comparison, Vd substantially reduced failure rates to 23.26%, 27.41%, and 20.82%, while increasing mean contrast ratios to 3.83, 4.19, and 3.91. Mean deficits likewise dropped from 1.36, 0.91, and 1.27 under Vf to 0.17, 0.29, and 0.15 under Vd. Our audit confirms that dynamic color assignment produced stronger contrast behavior than the fixed-red baseline in the tested route.

5.3 Measured Readability

Table 3 summarizes the paired t-test results for the *PREVis* scale, comparing the fixed-red baseline and dynamic-color condition across the three health visualization designs. For B **Fully-Integrated**, Vd received lower scores than Vf on all four PREVis subscales: *Understand*, *Data Readability*, *Layout*, and *Data Features*. All differences remained significant after Bonferroni correction ($p_{bonf} < .001$), with large effect sizes ($d = 1.30$–1.71), indicating a strong decrease in perceived readability and interpretability under the current dynamic-color condition. For A **Non-Integrated**, no subscale remained significant after correction. The strongest difference appeared for *Layout*, $t(23) = 2.945$, $p = .007$, $p_{bonf} = .084$, with lower ratings under Vd ($M = 5.35$) than Vf ($M = 5.99$) and a moderate effect size ($d = 0.60$). Other subscales showed no significant differences: *Understand* ($p = .09$, $d = 0.36$), *Data Readability* ($p = .08$, $d = 0.37$), and *Data Features* ($p = .387$, $d = 0.18$). For C **Partially-Matched**, no PREVis subscale differed significantly after correction. Effects were small to moderate for *Understand* ($p = .176$, $d = 0.29$), *Data Readability* ($p = .155$, $d = 0.30$), *Layout* ($p = .074$, $p_{bonf} = .888$, $d = 0.38$), and *Data Features* ($p = .092$, $d = 0.36$).

Overall, the current dynamic-color condition did not improve perceived readability over the fixed-red baseline. Its strongest negative effect appeared in the Fully-Integrated design, where all *PREVis* subscales decreased significantly; the other two designs showed only weak to moderate, non-significant differences after correction.

Table 3: Paired t-test results for the *PREVis* readability scale comparing Vf and Vd across three health visualization designs. Bonferroni-corrected p-values are reported as p_{bonf}, with Cohen's d as the effect size.

Design	Subscale	Mean Vf	Mean Vd	Mean Diff & 95% CI	t	df	p	p_{bonf}	d	Sig
A Non-Integrated	Understand	6.1528	5.8194	0.33 [−0.06, 0.72]	1.771	23	.09	1	0.36	No
	DataRead	5.9583	5.5833	0.38 [−0.05, 0.80]	1.830	23	.08	.96	0.37	No
	Layout	5.9861	5.3472	0.64 [0.19, 1.09]	2.945	23	.007	.084	0.60	No
	DataFeat	5.6042	5.4167	0.19 [-0.25, 0.63]	0.883	23	.387	1	0.18	No
B Fully-Integrated	Understand	6.2083	3.1944	3.01 [2.07, 3.95]	6.629	23	$< .001$	$< .001$	1.36	Yes
	DataRead	5.9306	2.7639	3.17 [2.39, 3.95]	8.388	23	$< .001$	$< .001$	1.71	Yes
	Layout	5.5000	3.1389	2.36 [1.59, 3.13]	6.355	23	$< .001$	$< .001$	1.30	Yes
	DataFeat	5.8542	2.7292	3.12 [2.28, 3.97]	7.691	23	$< .001$	$< .001$	1.57	Yes
C Partially-Matched	Understand	5.9306	5.5694	0.36 [−0.17, 0.90]	1.397	23	.176	1	0.29	No
	DataRead	5.8194	5.4167	0.40 [−0.16, 0.97]	1.472	23	.155	1	0.30	No
	Layout	5.4167	4.8472	0.57 [−0.06, 1.20]	1.869	23	.074	.888	0.38	No
	DataFeat	5.7917	5.3958	0.40 [−0.07, 0.86]	1.758	23	.092	1	0.36	No

5.4 Measured Aesthetics

Table 4 summarizes the paired t-test results for the *BeauVis* scale, comparing Vf and Vd across the three health visualization designs. The table reports Bonferroni-corrected p-values and Cohen's d for the three pairwise comparisons. For B **Fully-Integrated**, Vd received significantly lower aesthetic ratings ($M = 3.05$) than Vf ($M = 5.53$), $t(23) = 5.89$, $p < .001$, $p_{bonf} < .001$, with a large effect size ($d = 1.20$). For C **Partially-Matched**, ratings de-

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creased from Vf ($M = 5.30$) to Vd ($M = 4.78$), $t(23) = 2.11$, $p = .046$, but did not remain significant after correction ($p_{bonf} = .138$; $d = 0.43$). For A **Non-Integrated**, no significant difference was found, $t(23) = 1.96$, $p = .062$, $p_{bonf} = .186$, though Vd received a lower mean rating ($M = 5.15$) than Vf ($M = 5.59$; $d = 0.40$).

Overall, the current dynamic-color condition did not improve perceived aesthetics over the fixed-red baseline. The strongest negative effect appeared in the Fully-Integrated design, while the Partially-Matched and Non-Integrated designs showed weaker, non-significant differences after correction.

Table 4: Paired t-test results for the *BeauVis* aesthetics scale comparing Vf and Vd across three health visualization designs. Bonferroni-corrected p-values are reported as p_{bonf}, with Cohen's d as the effect size.

Design	Mean Vf	Mean Vd	Mean Diff & 95% CI	t	df	p	p_{bonf}	d	Sig
A Non-Integrated	5.5917	5.1500	0.44 [−0.24, 0.91]	1.961	23	.062	.186	0.40	No
B Fully-Integrated	5.5333	3.0500	2.48 [1.61, 3.36]	5.887	23	< .001	< .001	1.20	Yes
C Partially-Matched	5.3000	4.7750	0.52 [0.01, 1.04]	2.106	23	.046	.138	0.43	No

5.5 Objective Gameplay Performance

Table 5 summarizes objective gameplay measures from participants' logs, including hit and kill events, correct-hit efficiency, wrong-hit rate, correct-kill efficiency, and wrong-kill rate. These measures complement subjective ratings by capturing target-selection accuracy during play. Overall, the logs show no general performance advantage for dynamic color assignment. For B **Fully-Integrated**, Vd showed the weakest performance among all six conditions, with lower correct-hit efficiency (0.8353) and correct-kill efficiency (0.7637), and higher wrong-hit rate (0.1647) and wrong-kill rate (0.2363), than Vf. Conversely, Vf in this design showed the strongest overall performance, with correct-hit and correct-kill efficiencies of 0.9782 and 0.9694. For A **Non-Integrated**, performance was similar across color conditions, though Vf was slightly better in both hit and kill accuracy. For C **Partially-Matched**, performance was also comparable: Vd slightly improved correct-kill efficiency, while Vf was marginally better in correct-hit efficiency.

Together, the gameplay performance align with the subjective results: dynamic color assignment did not have clear improvement, with its weakest outcome in the Fully-Integrated design, where it was also rated less readable and less aesthetically pleasing.

5.6 Rankings

A **Non-Integrated** showed the strongest functional performance. As shown in Table 6, it received the most first-place rankings for *Readability* and *Mission Support*, suggesting that participants found the traditional bar design easier to read and more useful for identifying enemies. B **Fully-Integrated** showed weaker functional performance. Although it received moderate *Aesthetics* rankings, it was most often ranked third for *Readability* and *Mission Support*, suggesting that body-integrated health visualization may improve visual coherence but reduce mission-critical legibility. C **Partially-Matched** showed the strongest aesthetic performance and a more balanced result. It received the most first-place rankings for *Aesthetics* and many second-place rankings for *Mission Support* and *Positive Experience*, suggesting a better balance between visual integration and usability.

Overall, the rankings indicate distinct design strengths: A best supported readability and gameplay tasks, B improved integration at the cost of functional clarity, and C offered the strongest aesthetic appeal with acceptable usability.

Table 5: Condition-level objective gameplay performance derived from gameplay logs.

Condition	Total Hit Events	Correct-hit	Wrong-hit	Total Kill Events	Correct-kill	Wrong-kill
Vf Non-Integrated	4442	0.9458	0.0542	529	0.9314	0.0686
Vf Fully-Integrated	5436	0.9782	0.0218	642	0.9694	0.0306
Vf Partially-Matched	4233	0.9336	0.0664	508	0.9099	0.0901
Vd Non-Integrated	4273	0.9301	0.0699	516	0.9064	0.0936
Vd Fully-Integrated	2991	0.8353	0.1647	377	0.7637	0.2363
Vd Partially-Matched	4171	0.9419	0.0581	501	0.9226	0.0774

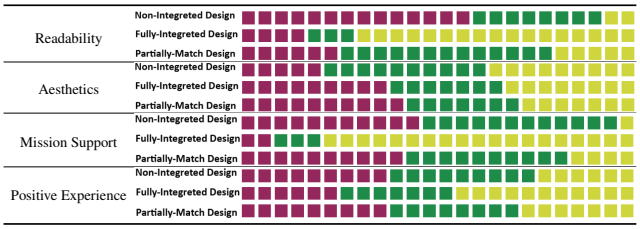


Table 6: Ranking Readability, Aesthetics, Mission Support and Positive Experience for 3 visualization types. : 1st place, : 2nd place, : 3rd place.

5.7 Thematic Analysis

We conducted post-interviews with 24 participants and identified seven themes as below. Details are in Table 7.

Overall Impression: Participants recognized the purpose of Vd dynamic color adaptation, but many still relied on the familiar Vf fixed-red baseline. P3, P10, P19, and P24 noted that fixed red enabled immediate recognition because it matched the common "red equals health" mental model. In contrast, three participants found adaptive colors chaotic or illogical, suggesting that for core status indicators, consistency may be more important than optimized visibility. P20 also noted that highly saturated red could cause false positives, as bright red objects might be mistaken for enemies.

Readability: Readability was closely tied to visualization design. While P11 found even A Non-Integrated difficult to interpret, many participants (P1—P3, P7—P12, P14—P16, P22) found B Fully-Integrated non-intuitive, especially in high-contrast scenes. Its unclear boundaries made health values hard to read. Four participants also found C Partially-Matched difficult to judge due to limited reference cues, especially near the 66% threshold.

Attention and Gameplay Flow: Some participants (P5, P6, P8, P13, P15, P24) felt the visualizations integrated smoothly without interrupting gameplay. However, three reported that Vd drew involuntary attention and made them double-check the interface. Another eight noted that B disrupted gameplay because its unclear shapes shifted attention from combat to reading health status.

Preference and Comparison: Participants generally prioritized fast information retrieval over visual integration. P1, P2, P13, P14, P16, P19, P20, and P24 preferred A because it was familiar and easy to read. Five participants felt B reduced character visual quality by making robots look unnatural. P19 and P23 also noted that C could be occluded from high vertical camera angles.

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Table 7: Themes, Sub-Themes, Representative Quotes, and Participants

Themes	Sub-Themes	Representative Quotes	Participants
Overall Impression / Review	Preference for a Fixed Red Color Scheme	When it comes to color, the red one definitely helps me identify the health bar more easily.	P3, P10, P19, P24
	Perceptions of Color Adaptation	The color changes make it feel chaotic and illogical to me. In fact, I think there is no need for the color to change at all.	P10, P16, P20
	Perceived Ineffectiveness of the Fixed Red Color	If the color becomes extremely red, I tend to subjectively perceive it as representing an enemy.	P20
Readability	Difficulty in Assessing Health Values Across Visualization Type	The bar-type health indicator (Non-Integrated) is actually quite difficult to interpret as well.	P11
		I find the fully-integrated health display is not very intuitive, especially in high-contrast scenes where the health value is not very apparent.	P1–P3, P7–P12, P14–P16, P22
		For the circular type (partially-matched), it is really difficult to distinguish the critical value of 66%. The bar-type (Non-Integrated) and Fully-Integrated indicators have clearer reference points, but the circular one lacks such cues.	P3, P19, P20, P23
	Value-specific Readability Challenges	The 66% value is not intuitive. Normally, the remaining health on a traditional bar is quite clear, but with this setting, it becomes less straightforward.	P5, P8, P10, P12, P14, P17
	No Impact from Color Adaptation	I think it's fine; it doesn't affect me much.	P6, P11, P12, P15, P20, P23, P24
Attention & Gameplay Flow	Experiences of Distraction	When I'm continuously playing, if I see something flashing or changing on the right side, I get distracted and glance at it. Sometimes I even double-check and look at it a couple more times.	P19, P20, P21
	Interference with Gameplay Flow	For the Fully-Integrated type, I couldn't read the color at all, so I couldn't really play. Even after it was changed to red, because the shape is irregular, it makes reading the percentage quite difficult. I often focus on reading it carefully instead of actually fighting robots.	P2, P4, P7, P9, P10, P16, P17, P20
	No Influence on Gameplay Flow	There was no interference with the gameplay flow; I didn't feel any impact.	P5, P6, P8, P13, P15, P24
Preference & Comparison	Prefer / Dislike Non-Integrated	If I need to quickly and intuitively obtain this information, I may prefer the bar-type (Non-Integrated).	P1, P2, P13, P14, P16, P19, P20, P24
		The bar-type (Non-Integrated) is actually quite difficult to interpret as well.	P11
	Prefer / Dislike Fully-Integrated	If it's a more immersive experience, I might prefer the type where the color on the robot's body changes (Fully-Integrated).	P5, P6, P8, P16, P19
		What I dislike is the fully-rendered body (Fully-Integrated). Normally, a character model looks aesthetically pleasing, but once the health color is rendered onto it, the character's appearance is definitely affected in a negative way.	P12, P14, P15, P16, P24
	Prefer / Dislike Partially-Matched	The circular indicator (Partially-Match) fits better with the robot.	P17, P18, P22
		When I am at a higher position, the circular indicator below faces downward and is not aligned with my main viewing angle. I need to move down to clearly see whether the health is above 66%.	P19, P23
Color Adaptation Experience	Confusion and Difficulties Regarding Dynamic Colors	The main impact comes from my habit of treating it as a health bar. When the health bar changes color, I wonder if it represents another function or indicates a different character identity.	P10, P14, P16, P22
		With the Fully-Integrated type, I encountered a problem during gameplay: the color on the character's body often overlaps with the environment or lighting, making it harder to distinguish.	P1, P10, P11, P12, P14, P16, P17, P18, P19
	No Perceived Effect from Dynamic Colors	I didn't feel there was much impact.	P2, P6, P11, P12, P15, P20, P23, P24
	Positive Evaluations of Dynamic Colors	As long as the contrast is high, I think it is relatively easy to see.	P4, P5, P9, P14
Visual Fatigue	Visual Fatigue Induced by Gameplay	The movement speed in this game is very fast, including jumping and so on, so I felt a bit dizzy, especially towards the end of the session.	P3, P6, P8, P11, P15, P19
	Visual Fatigue Attributed to Color	Watching dynamic colors is more straining on the eyes and can be tiring. As someone who is a bit color weak, I find it somewhat difficult to distinguish.	P7, P16, P21, P22
		I personally haven't experienced it, but I can imagine that if you play for a longer period of time, the bright red color might have a slight impact.	P10
	Absence of Visual Fatigue	No, because the overall scene is very simple. I think just focusing on the monster does not cause any visual fatigue.	P2, P4, P5, P8, P12, P13, P18, P20, P23, P24
Improvement Suggestions	Suggestions for Color Algorithm	The color saturation could be adjusted according to the amount of health.	P6
		Do not use colors that are too similar to the environment or to the character's model.	P2, P3, P4, P9
	Suggestions for Improving Visualization	There could be a dedicated reference line at 66%. Having an additional guideline makes it easier to judge.	P3, P12
		Adding particle or motion effects may make the health bar more distinguishable.	P15, P18

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Color Adaptation Experience: Participants' responses suggest that meeting contrast requirements does not necessarily improve experience. P4, P5, P9, and P14 felt Vd supported perception when contrast stayed high. However, P10, P14, P16, and P22 interpreted color changes as game states, enemy types, or special effects, indicating semantic interference from breaking the red-health mapping. Ten participants also reported that B adaptive colors sometimes blended with lighting or robot textures, making them harder to distinguish than Vf. Thus, adaptation should account for object texture and gameplay semantics, not just background contrast.

Visual Fatigue: Ten participants reported little or no visual fatigue, partly because the scenes were simple and the sessions were short. Six participants attributed dizziness mainly to game movement, such as jumping and spinning. However, four participants (P7, P16, P21, P22), including some with weak color vision, found Vd dynamic colors more tiring than Vf fixed red, suggesting that continuous chromatic changes can increase perceptual load.

Improvement Suggestions: Participants suggested combining color adaptation with clearer geometric cues. P6 suggested linking saturation to health values, while P2, P3, P4, and P9 warned against colors close to the environment. P3 and P12 recommended adding a 66% reference line, and P15 and P18 suggested particle effects, motion, or texture cues as alternatives to hue changes.

Overall, the analysis highlights trade-offs among visibility, readability, and aesthetics. Although Vd can improve contrast, it may disrupt the familiar "red equals health" convention and increase cognitive load. Participants' preference for A suggests that clear geometric structure is often more valuable than aesthetic integration in fast-paced environments, and that color adaptation alone is insufficient for precise and comfortable data identification.

6 Discussion and Limitations

Our findings show that adaptive color design in fast-paced games is not only a contrast-optimization problem, but also a perceptual and semantic one: colors must remain visible, interpretable, and consistent with players' gameplay expectations.

6.1 Algorithmic Constraints and Color Model Selection

A key challenge is how the algorithm represents the surrounding visual context. We estimated background color by averaging pixels within a sampling area. This is computationally efficient, but can be misleading when the background contains multiple colors, textures, or high-contrast patterns. In such cases, the mean color may not match any actual visible region, leading to mathematically reasonable but visually suboptimal choices.

The use of HSV also reflects a trade-off between real-time performance and perceptual accuracy. HSV supports fast adjustments to Saturation and Value while preserving Hue, which helps reduce latency when multiple moving enemies are updated at high frame rates. However, HSV is not perceptually uniform: equal numerical changes do not always produce equal perceptual changes. For example, saturated yellow can appear much brighter than saturated blue despite comparable HSV values. Although CIELAB offers stronger perceptual uniformity, it is more computationally expensive for frame-by-frame updates on multiple targets. Direct RGB adjustment is also less intuitive when preserving hue while improving visibility. Thus, HSV was a practical real-time compromise, but its perceptual limitations help explain why optimized colors did not always improve readability.

6.2 Impact on User Experience and Cognitive Load

Improved contrast alone did not guarantee better user experience. In our gray-green scenes, the algorithm often produced a narrow range of "safe" adaptive colors. In B Fully-Integrated, these colors sometimes blended with the robot's texture, lighting, or shadows, and the irregular body geometry further reduced health-value legibility. This suggests that adaptive color design should consider object surface, shape, and visual complexity, not only sampled-background contrast. Our comparison also does not isolate temporal adaptation alone: Vd changed color over time, altered contrast behavior, and disrupted the red-health mapping preserved in Vf. Thus, the null or negative effects reflect the combined impact of adaptive hue changes, contrast adjustment, and semantic disruption. Dynamic color also disrupted players' mental models. Because red is commonly associated with health, damage, or danger, participants could interpret it quickly during gameplay. When Vd changed hue, some participants read the changes as new game states, enemy types, shields, or special effects, introducing semantic ambiguity and extra interpretation work. This matches our interviews: many participants relied on red as an immediate health cue, while adaptive hues were sometimes interpreted as game-state or special-effect indicators.

This ambiguity affected gameplay flow: changing colors sometimes drew attention and made players re-check the interface, competing with the combat task. The system also offered limited accessibility benefits, as some participants with weak color vision found continuous chromatic changes tiring or harder to track than a stable baseline. Overall, adaptive color systems should avoid frequent semantic changes and prioritize stable, learnable mappings.

6.3 Limitations and Design Considerations

Our results should be interpreted within the tested environment. The study used a single *RobotLife* map, which limits generalizability to environments with different color distributions, lighting, textures, and visual complexity. The route-based contrast audit showed that Vd substantially improved contrast over Vf in the tested route. Thus, our findings do not show that dynamic color failed to improve contrast; rather, they show that, in this environment and implementation, improved contrast alone did not lead to better perceived readability or aesthetics. Future work should test adaptive color design across more gameplay environments and visual conditions.

Our findings suggest several design considerations. First, adaptive systems should avoid frequent hue changes for semantically important information; preserving red while adjusting saturation, brightness, outline strength, or glow intensity may improve visibility without breaking the red-health convention. Second, integrated designs need clearer geometric references, such as a fixed 66% marker, clear boundaries, or outer contours, especially on irregular surfaces. Third, segmented indicators could help players estimate health by counting blocks. Finally, adaptive algorithms should consider background contrast together with object texture, lighting, camera angle, and gameplay semantics to improve visibility while preserving readability, attention, and player expectations.

7 Conclusions

We presented a real-time, background-adaptive color assignment approach for in-game visualizations and evaluated it with 24 participants. In *RobotLife* [3], our dynamic color assignment approach did not significantly improve readability or aesthetics during fast-moving gameplay. Our findings suggest that color for in-game visualizations acts as both a perceptual channel and a semantic cue: participants relied on fixed-color conventions for quick interpretation, while dynamic color assignments introduced confusion, distraction, and cognitive load. Thus, adaptive color should be used cautiously for semantically important game information and paired with stable visual structures, clear geometric cues, and consistent color semantics.

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Supplemental Material Pointers

All supplemental materials for this work are available on OSF at osf.io/csjrb/. Respectively, we share our (a) source code and a demo of the game; (b) user study materials and raw data, including the pre-, in-game, and post-game questionnaires, interview questions, as well as participants' original responses and interview records; and (c) analysis results, specifically the quantitative data results and the thematic analysis of the interviews.

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